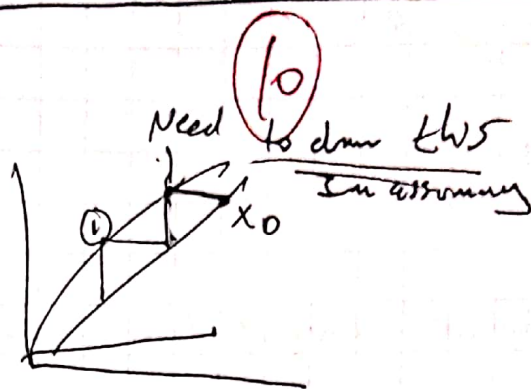
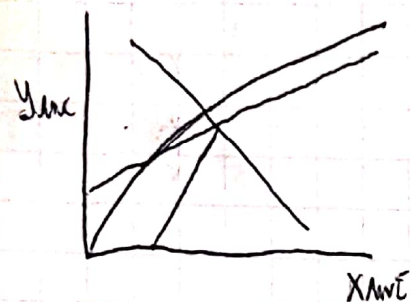


A2) McCabe Thiele diagram

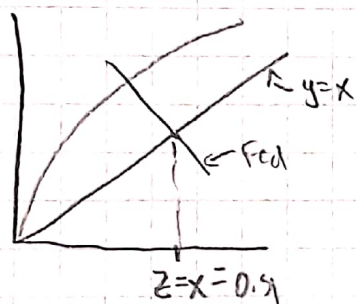


a. (1) Actual feed tray?

↳ Switch from top OP → Bottom of 1st

⇒ 6

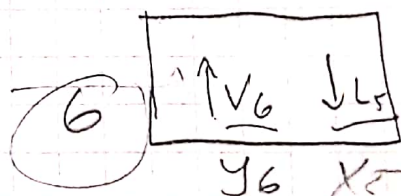
(2)  $z_{mvc} = ?$



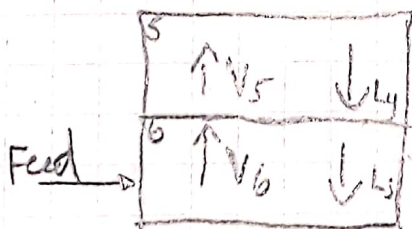
$z_{mvc} = 0.9$

(3) Vapor composition on Feed tray

Feed Tray



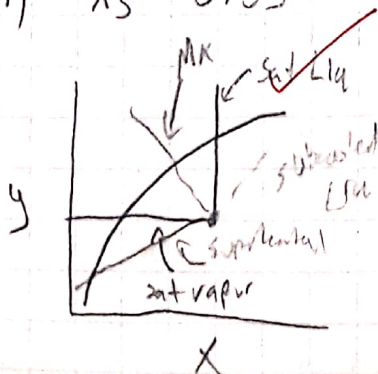
COM



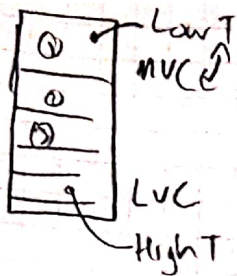
graph  
 $\Rightarrow V_6 \rightarrow y_6 = 0.52$   
 $L_5 \rightarrow x_5 = 0.33$

(4)  $x_5 = 0.33$

b.



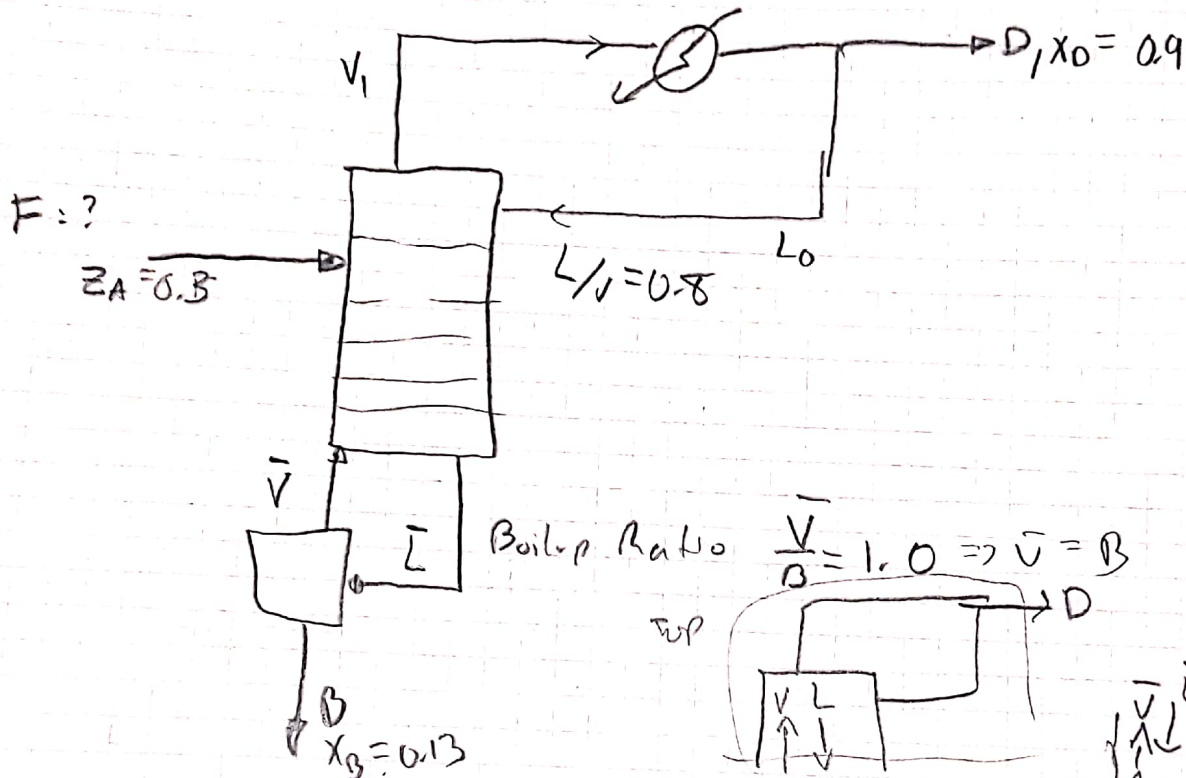
⇒ 2-phase feed



NOTE MVC boils at lower Temp.  
 Stage 1 has higher conc. of MVC  
 $\Rightarrow$  Stage 7 Higher temp than stage 1

D.7) Distillation column w/ total condenser  
 Acetone & ethanol

(MO)



TOP  
 $V = L + D \Rightarrow D = V - L$

$$Vy = Lx + Dx_D$$

$$y = \frac{L}{V}x + \frac{D}{V}x_D$$

$$y = \frac{L}{V}x + \frac{V-L}{V}x_D$$

$$\frac{L}{V} = 0.8 \Rightarrow y = \frac{0.8}{1}x + (1 - \frac{0.8}{1})x_D$$

$$\Rightarrow y = 0.8x + (1 - 0.8)x_D$$

$$x_D = 0.9 \Rightarrow y = 0.8x + 0.2x_D$$

TOP  
 $\Rightarrow y = 0.8x + 0.18$

Bottom

$$L = V + B \Rightarrow$$

$$Lx = Vy + Bx_B$$

$$\frac{V}{B} = 1 \Rightarrow y = \frac{L}{V}x - \frac{B}{V}x_B$$

$$\Rightarrow y = \frac{(2V)}{V}x - (1)x_B$$

Bottom  
 $y = 2x - x_B$

$$x_B = 0.13 \Rightarrow y = 2x - 0.13$$

Note:  $\left(\frac{V}{B} = 1\right) \Rightarrow V = B$   
 $L = V + B = V + V$   
 $(L = 2V)$

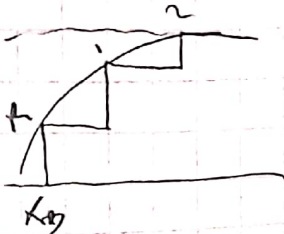


4. 5th stage comp  $\rightarrow X_5 = ?$



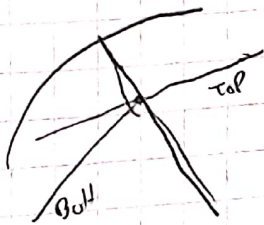
$$X_5 = 0.52$$

b.  $y_2$  above reboiler



$$y = 0.52$$

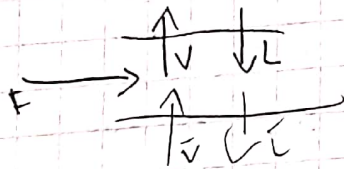
c. Optimum feed location



feed plate location  $\sim 8$

# stages  $\sim 8 + h$

q feed  $\sim 0.67$



$$F + \bar{V} + L = V + \bar{L}$$

$$Fz + \bar{V}y + Lx = Vy + \bar{L}x$$

$$Vy - \bar{V}y = Lx - \bar{L}x + Fz$$

$$y(V - \bar{V}) = (L - \bar{L})x + Fz$$

$$y = \frac{L - \bar{L}}{V - \bar{V}}x + \frac{Fz}{V - \bar{V}}$$

$$y = \frac{LF}{VF}x + \frac{F}{VF}z$$

$$y = -\left(\frac{LF}{F - LF}\right)\frac{F}{F_x} + \frac{F}{F - LF}z$$

$$y = -\frac{LF/F}{F/F - LF/F} + \frac{F/F}{F/F - LF/F}z$$

$$y = -\frac{q}{1 - q}x + \frac{1}{1 - q}z$$

$$\text{Slope: } \frac{\text{rise}}{\text{run}} = \frac{0.35 - 0.3}{0.76 - 0.5} = \frac{-0.05}{0.04} = -1.25$$

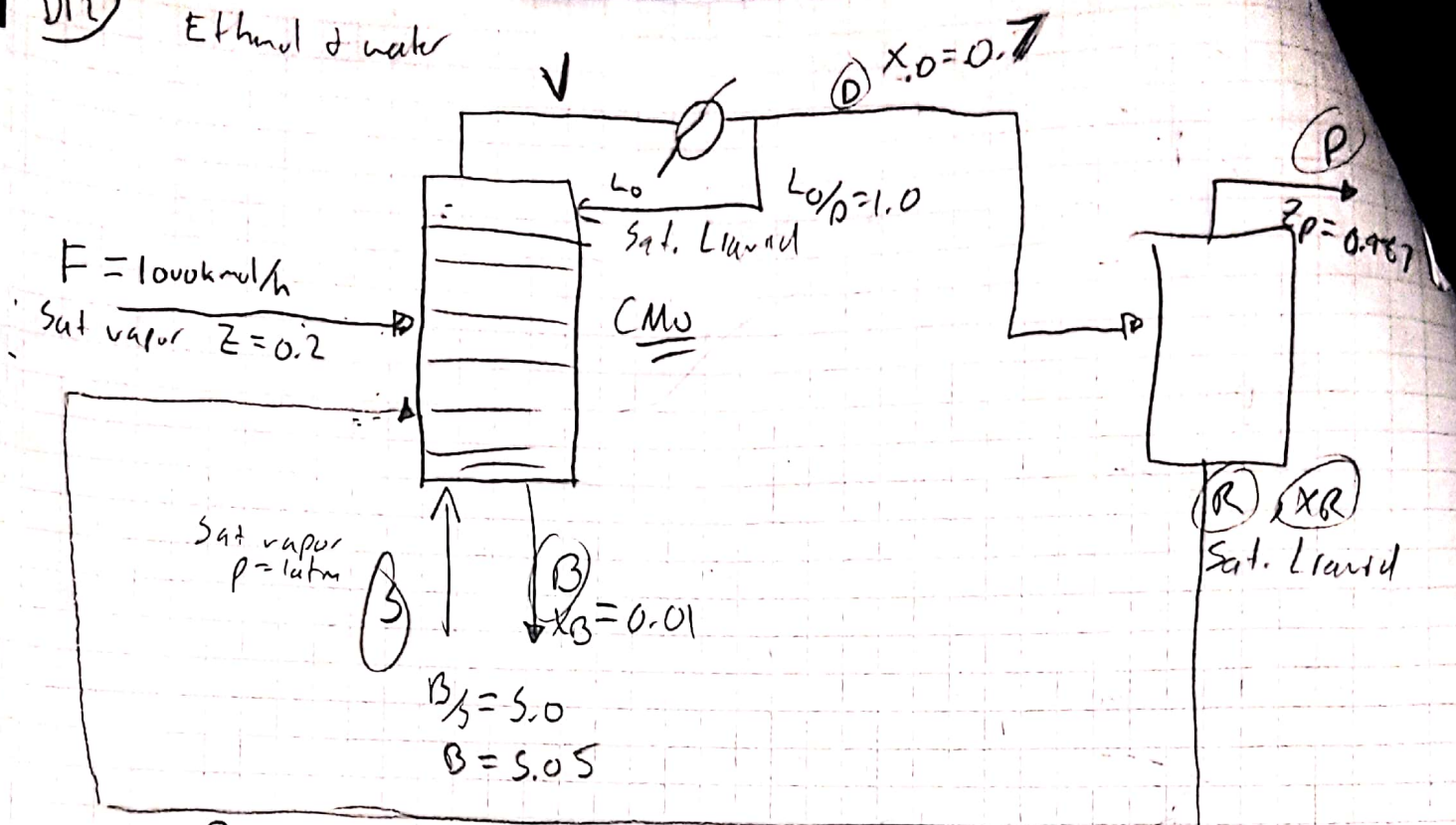
$$-1.25 = -\frac{q}{1 - q}$$

$$1 - 1.25q = 1 - q$$

$$1 = 1.25q$$

$$q = 1/1.25 = 0.8$$

Dist Ethanol & water



Overall:  $F + S = B + P$   $F(1-z) + S = B(1-x_B) - P(1-z_P)$

Overall Ethanol:  $Fz + Sx_0 = Bx_B + Pz_P$

Separation:  $P + R = D$   
Overall:  $F + S = B + P$

$F + S = (55) + P$

$F - 45 = P$

$P = (1000 - 4(201.898))$

$P = 192.406 \text{ kmol/h}$

$B = 55 = 5(201.898)$

$B = 1009.442 \text{ kmol/h}$

Separation:  $D = P + R$

MVL:  $Dx_D = Pz_P + R x_R$

$Fz + 0 = Bx_B + Pz_P$

$Fz = (55)x_B + (F - 45)z_P$

$Fz = 55x_B - 45z_P + Fz_P$

$Fz - Fz_P = 55x_B - 45z_P$

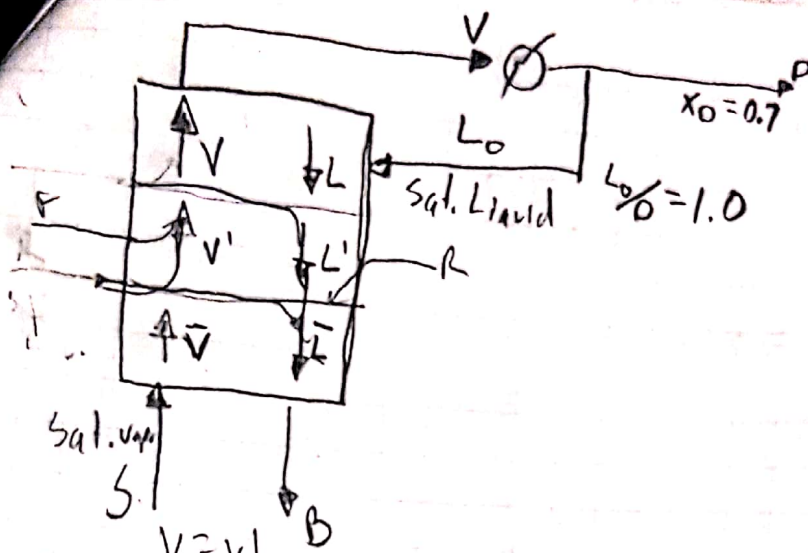
$S = \frac{(Fz - Fz_P)}{(5x_B - 4z_P)} = \frac{(1000 \frac{\text{kmol}}{\text{h}})(0.2) - (1000 \frac{\text{kmol}}{\text{h}})(0.2)}{5(0.01) - 4(0.987)}$

$S = 201.898 \text{ kmol/h}$

NOTE

$\frac{B}{S} = 5$   
 $B = 5S$





Top Vapor:  $V = V'$

Top Liquid:  $L = L_0$

Top Vapor:  $V' = F + \bar{V}$

Top Vapor:  $V' = 1000 + 201.898 = 1201.898$

Bottom Vapor:  $\bar{V} = \bar{V} + F$

Bottom Liquid:  $L' = L + L_0 \Rightarrow (600 + 408.543) = 1009$

Bottom Vapor:  $\bar{V} = S = 201.898 \text{ kmol/h}$

Bottom Liquid:  $\bar{L} = B = 1009.492$

①  $F + V' = V$

$F + S = V$

③  $L' + R = \bar{L} = B$

$V = 1000 + 201.898$

$V = 1201.898 \text{ kmol/h}$

$R = B - L'$

$R = B - L_0$

$= 1009.492 - 600.949$

$R = 408.543 \text{ kmol/h}$

②  $V = L_0 + D$  NOTE  $\frac{L_0}{D} = 1.0$

$V = 2L_0 = 1201.898$

$L_0 = V/2 = \frac{1201.898}{2}$

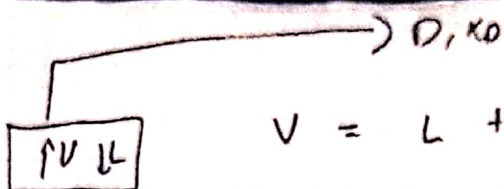
$L_0 = 600.949 \text{ kmol/h}$

SAP:  $P_{XD} = P_{ZF} + R_{XR}$

$X_R = \frac{P_{XD} - P_{ZF}}{R} = \frac{(600.949)(0.7) - (192.406)(0.981)}{408.543}$

$X_R = 0.5648$

Top



$$V = L + D \Rightarrow D = V - L$$

$$V y = L x + D x_D$$

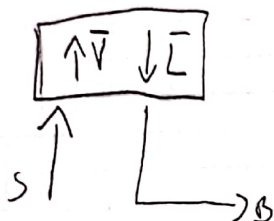
$$y = \frac{L}{V} x + \frac{D}{V} x_D$$

$$y = \frac{L}{V} x + \left(1 - \frac{L}{V}\right) x_D$$

$$y = \frac{600.949}{1201.898} x + \left(1 - \frac{600.949}{1201.898}\right) 0.7$$

$$[y = 0.5 x + 0.35] \text{ TOP}$$

Bottom



$$S + L = V + B$$

$$S x + L x = V y + B x_B$$

$$y = \frac{L}{V} x - \frac{B}{V} x_B$$

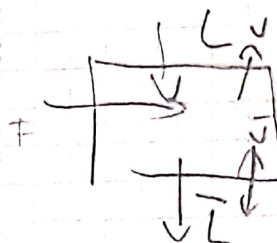
$$y = \frac{1009.492}{201.898} x - \frac{1009.492}{201.898} (0.01)$$

$$[y = 5 x - 0.05] \text{ Bottom}$$

Feed Line sat vapo  $\Rightarrow q = 0$

$$y = \frac{q}{q-1} x + \frac{1}{1-q} z$$

$$y = z = 0.2$$



Recycle feed sat Liq.  $q = 1$

$$y = \frac{q}{q-1} x + \frac{1}{1-q} z$$

$$y = \frac{1}{0} x + \frac{1}{0} z$$

Infinite line

$$\Rightarrow x = z$$

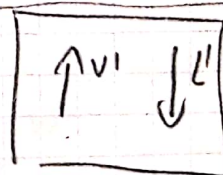
Co-hexcycle steam  $\sim 1$   
Feed steam  $\sim 4$   
# of eq stages  $\sim 5$

Msd OP

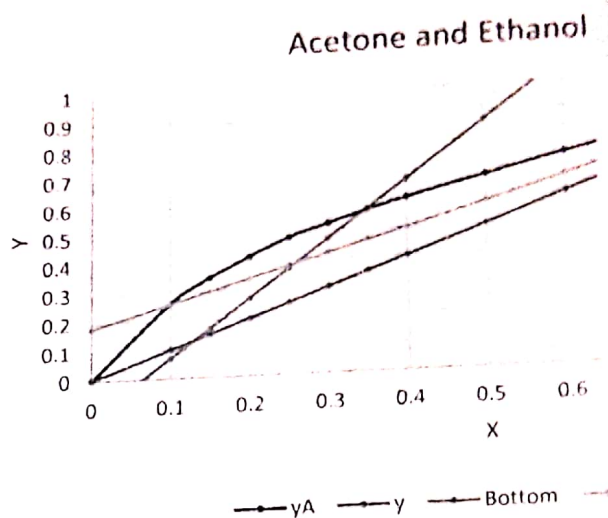
$$y = \frac{L'}{V'} x + \frac{D x_D - B x_B}{V'}$$

$$= \frac{600.949}{1201.898} x + \frac{(600.949/0.7) - (1008.949/0.9648)}{1201.898} F_2$$

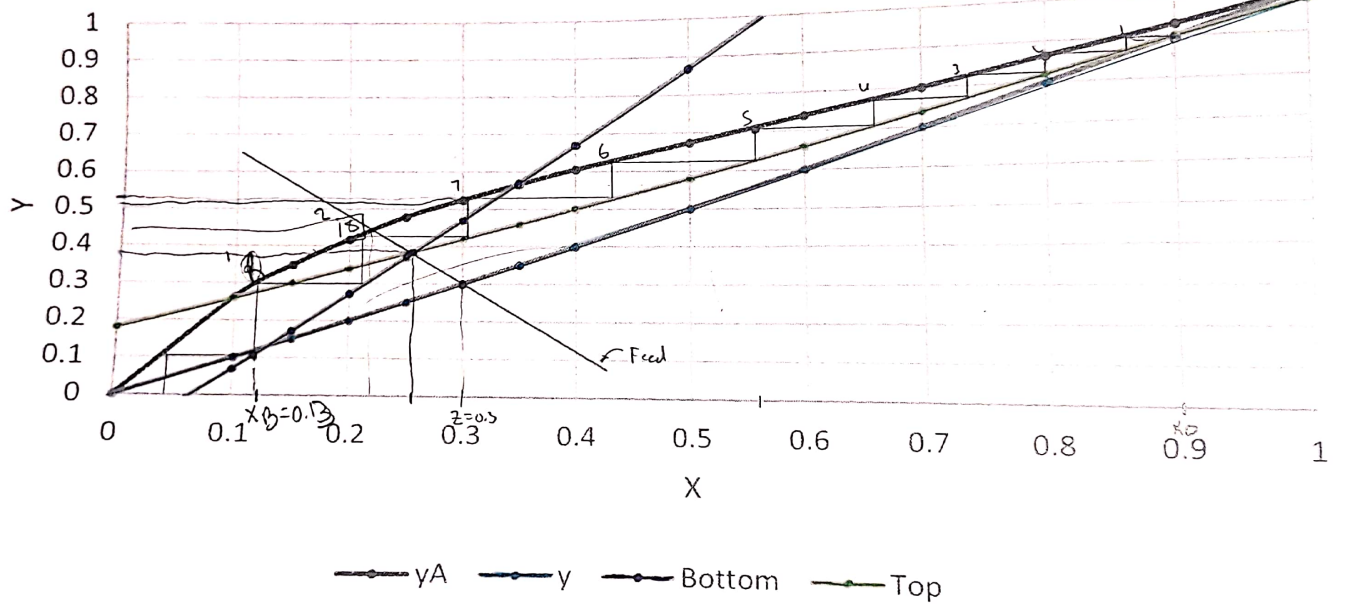
$$= 0.5349 x + 0.135$$



| $x_A$ | $y_A$ | $y$  | Bottom | Top  |
|-------|-------|------|--------|------|
| 0     | 0     | 0    | -0.13  | 0.18 |
| 0.1   | 0.262 | 0.1  | 0.07   | 0.26 |
| 0.15  | 0.348 | 0.15 | 0.17   | 0.3  |
| 0.2   | 0.417 | 0.2  | 0.27   | 0.34 |
| 0.25  | 0.478 | 0.25 | 0.37   | 0.38 |
| 0.3   | 0.524 | 0.3  | 0.47   | 0.42 |
| 0.35  | 0.566 | 0.35 | 0.57   | 0.46 |
| 0.4   | 0.605 | 0.4  | 0.67   | 0.5  |
| 0.5   | 0.674 | 0.5  | 0.87   | 0.58 |
| 0.6   | 0.739 | 0.6  | 1.07   | 0.66 |
| 0.7   | 0.8   | 0.7  | 1.27   | 0.74 |
| 0.8   | 0.865 | 0.8  | 1.47   | 0.82 |
| 0.9   | 0.929 | 0.9  | 1.67   | 0.9  |
| 1     | 1     | 1    | 1.87   | 0.98 |



# Acetone and Ethanol

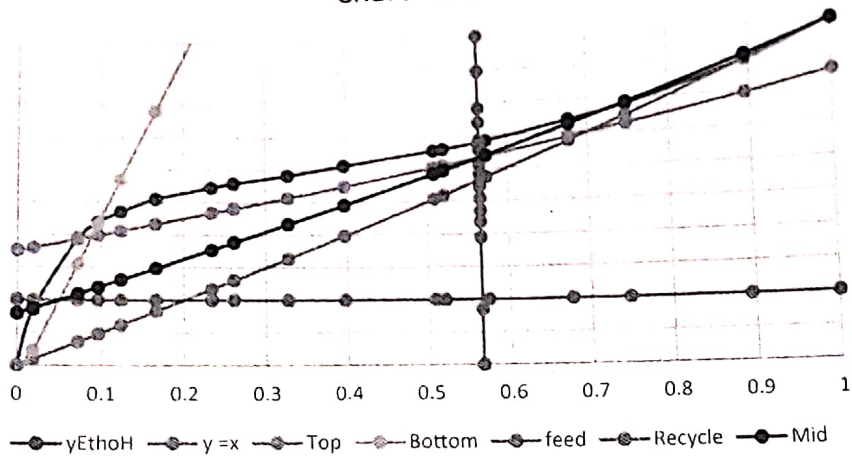




| x EthoH | yEthoH | y=x    | Top     | Bottom | x      | feed | Mid      |
|---------|--------|--------|---------|--------|--------|------|----------|
| 0       | 0      | 0      | 0.35    | -0.05  | 0.5648 | 0.2  | 0.158    |
| 0.019   | 0.17   | 0.019  | 0.3595  | 0.045  | 0.5648 | 0.2  | 0.173958 |
| 0.0721  | 0.3891 | 0.0721 | 0.38605 | 0.3105 | 0.5648 | 0.2  | 0.218557 |
| 0.0966  | 0.4375 | 0.0966 | 0.3983  | 0.433  | 0.5648 | 0.2  | 0.239134 |
| 0.1238  | 0.4704 | 0.1238 | 0.4119  | 0.569  | 0.5648 | 0.2  | 0.26198  |
| 0.1661  | 0.5089 | 0.1661 | 0.43305 | 0.7805 | 0.5648 | 0.2  | 0.297507 |
| 0.2337  | 0.5445 | 0.2337 | 0.46685 | 1.1185 | 0.5648 | 0.2  | 0.354285 |
| 0.2608  | 0.558  | 0.2608 | 0.4804  | 1.254  | 0.5648 | 0.2  | 0.377046 |
| 0.3273  | 0.5826 | 0.3273 | 0.51365 | 1.5865 | 0.5648 | 0.2  | 0.432899 |
| 0.3965  | 0.6122 | 0.3965 | 0.54825 | 1.9325 | 0.5648 | 0.2  | 0.49102  |
| 0.5079  | 0.6564 | 0.5079 | 0.60395 | 2.4895 | 0.5648 | 0.2  | 0.584585 |
| 0.5198  | 0.6599 | 0.5198 | 0.6099  | 2.549  | 0.5648 | 0.2  | 0.59458  |
| 0.5732  | 0.6841 | 0.5732 | 0.6366  | 2.816  | 0.5648 | 0.2  | 0.639431 |
| 0.6763  | 0.7385 | 0.6763 | 0.68815 | 3.3315 | 0.5648 | 0.2  | 0.726024 |
| 0.7472  | 0.7815 | 0.7472 | 0.7236  | 3.686  | 0.5648 | 0.2  | 0.785573 |
| 0.8943  | 0.8943 | 0.8943 | 0.79715 | 4.4215 | 0.5648 | 0.2  | 0.909123 |
| 1       | 1      | 1      | 0.85    | 4.95   | 0.5648 | 0.2  | 0.9979   |

1  
0.9  
0.8  
0.7  
0.6  
0.5  
0.4  
0.3  
0.2  
0.1  
0

Chart Title



D12

Chart Title

